

Global Trends in Religious Composition: Secularization and Socio-Economic Development

A Global Data-Driven Analysis

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Abstract

This report studies religious secularization — the share of a country's population reporting no religious affiliation — and asks how it relates to four well-known markers of socio-economic development: fertility rate, urbanization, life expectancy, and GDP per capita. The dataset combines the World Religion Project (WRP 2020) with the World Bank's World Development Indicators (WDI), covering 954 country-year observations from 1960 to 2010.

The analysis is organized into two parts. Part I (unsupervised learning) uses correlation analysis, Principal Component Analysis (PCA), k-means clustering, and hierarchical clustering to explore how countries group together. Part II (supervised learning) builds and compares several models of secularization: simple and multivariate OLS regression, Ridge and Lasso regression, and Random Forest regression, evaluated with 5-fold cross-validation.

The headline result, confirmed by every method used, is that fertility rate is the strongest and most reliable predictor of secularization. Income (GDP per capita), by contrast, is a weak and unreliable predictor once fertility is accounted for. Clustering further shows that the most secular group of countries in the sample is not the richest group — directly challenging the simple idea that “richer automatically means more secular.”

1. Problem Statement and Dataset

1.1 Goal of the Analysis

The central question this report asks is simple: which socio-economic and demographic factors best predict how religious, or non-religious, a country's population is? Secularization is measured here as the percentage of the population with no religious affiliation (the WRP variable `unaffiliated_pct`, relabelled secularization in this report) — a standard proxy used in comparative religion research.

In plain terms, the report tries to answer three things:

1. How strongly does secularization move together with fertility, urbanization, life expectancy, and income — and which of these matters most?
2. Does a flexible, non-linear model (Random Forest) predict secularization meaningfully better than a straight-line model (OLS), and is that improvement reliable rather than a fluke of one particular train/test split?
3. Do countries fall into a small number of natural “types” when grouped only by their development and secularization profile, without telling the algorithm which type each country belongs to?

1.2 Data Sources

Source	Variables Used	Coverage
World Religion Project (WRP 2020)	% unaffiliated, % Christian, % Muslim, etc.	165 countries, 1945–2010 (5-year intervals)
World Bank WDI	Fertility rate, urbanization, life expectancy, GDP per capita	Global, annual, 1960–2010
Correlates of War (COW)	Country code → country name lookup only	All countries

After merging WRP and WDI on country code and year (an inner join, keeping only rows present in both tables), the final dataset has 954 country-year observations across 25 variables. Secularization in this sample ranges from close to zero (in strongly religious societies) to over 70% (in a handful of strongly secular societies, mostly in Eastern Europe and East Asia).

Summarizing, the cleaning steps were: keep only the religion percentage columns from WRP, convert them from 0–1 proportions into 0–100 percentages, attach a readable country name using the COW lookup table, reshape the World Bank data from a wide year-by-year format into a tidy long-then-wide format, and finally drop any row missing a needed value before each specific analysis (so the exact sample size varies slightly from one model to the next, as noted throughout).

2. Key Findings

- 1. Fertility is the single strongest predictor of secularization**, in every model tried. Across simple OLS, multivariate OLS, Ridge, Lasso, and Random Forest, fertility rate consistently shows the strongest relationship with secularization, and **it is the only predictor whose effect survives essentially unchanged across all of these very different methods**. In plain terms: **knowing a country's birth rate tells you more about its religiosity than knowing its income, education, or urbanization level**.
- 2. Income (GDP per capita) has a weak and unreliable effect on secularization on its own**. The raw correlation between GDP and secularization is small ($r = 0.13$), and the GDP coefficient is **not statistically significant** in the multivariate OLS ($p = 0.55$). Even Lasso (a method specifically designed to discard unhelpful predictors), **shrinks the GDP coefficient close to zero before it shrinks fertility's**. This means that, by itself, **being a richer country does not reliably predict being a more secular one**.
- 3. Random Forest substantially outperforms linear regression**. A flexible, non-linear model substantially outperforms straight-line regression. Random Forest explains **56.1% of the variance in secularization** ($R^2 = 0.561$) on a single held-out test set, compared to only **20.5% for the best linear model**(multivariate OLS). Critically, this advantage **holds up under 5-fold cross-validation** (mean $R^2 = 0.591$, $SD = 0.036$, range=0.556 - 0.629) showing that **the improvement is a real and repeatable pattern in the data rather than a lucky train/test split**.
- 4. The fertility–secularization relationship is non-linear, it bends and does not run in a straight line**. The partial dependence plot shows secularization rising steeply as fertility falls from about **7.5 to 5 children per woman**, before gradually levelling off at lower fertility levels. A linear model cannot capture this shape, which is the primary reason why **Random Forest beats OLS so clearly** (Finding 3).
- 5. The most secular countries are not the richest countries. Four distinct country “types” emerge from the data, and the most secular group is not the richest group**. K-means clustering identifies:
 - A large **Traditional / High Fertility** cluster
 - A large **Developing Religious** cluster
 - A small **Highly Developed / Secular** cluster
 - A small **Secularizing Middle-Income** clusterThe most striking result is that the **Secularizing Middle-Income cluster has the highest average secularization (34.5%)**, despite having only moderate income levels. By contrast, the **Highly Developed / Secular cluster averages only 9.6% secularization**.

This is the strongest evidence in the report against the simple idea that “richer automatically means more secular.”

PART I — UNSUPERVISED LEARNING

3. Exploratory Data Analysis

3.1 Global Secularization Trend

As a first, simple look at the data, we average secularization across all available countries for each year and plot the trend over time.

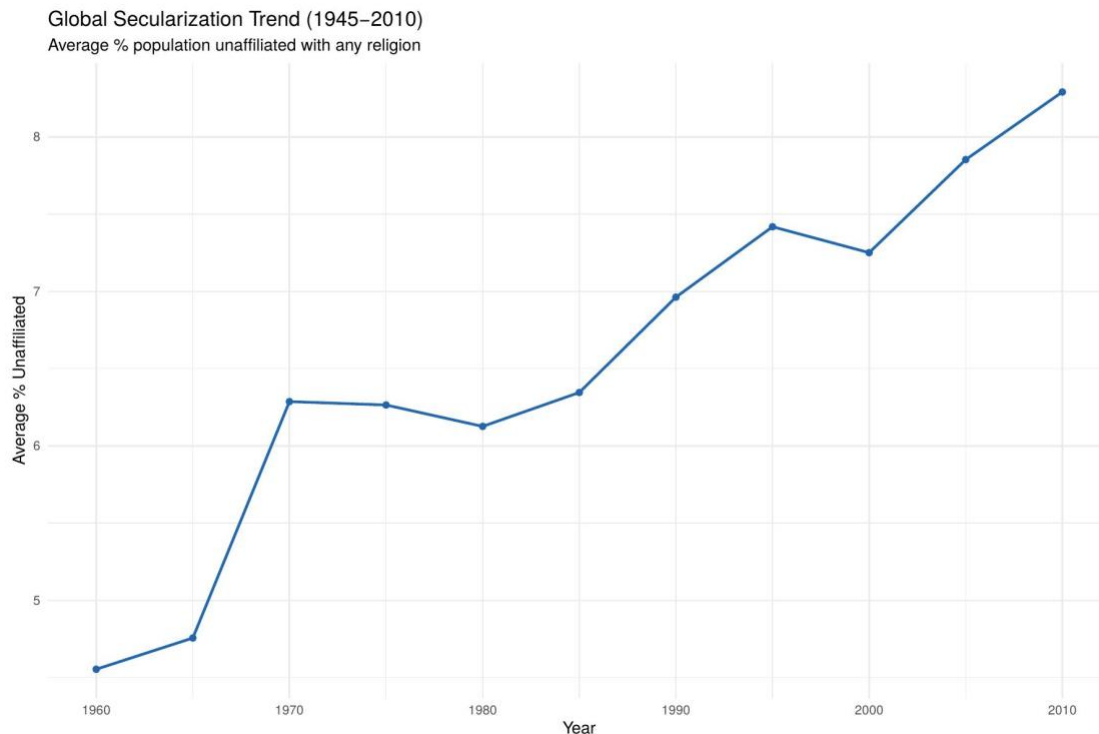


Figure 1. Average % of population unaffiliated with any religion, 1960–2010.

We observe that, the world-average secularization line moves between roughly 5% and 8% over this fifty-year window, without a single steady upward climb. This tells us secularization is not simply “everywhere increasing every year” — it is a story of some countries changing a lot and others barely changing, which averages out into a noisier global picture. This is exactly why the rest of the report looks across countries (cross-sectionally) rather than relying on the time trend alone.

3.2 Correlation Structure

Next, we measure how strongly each pair of variables moves together, using the Pearson correlation coefficient (a number between -1 and $+1$, where 0 means no relationship, $+1$ means they move perfectly together, and -1 means one rises exactly as the other falls).

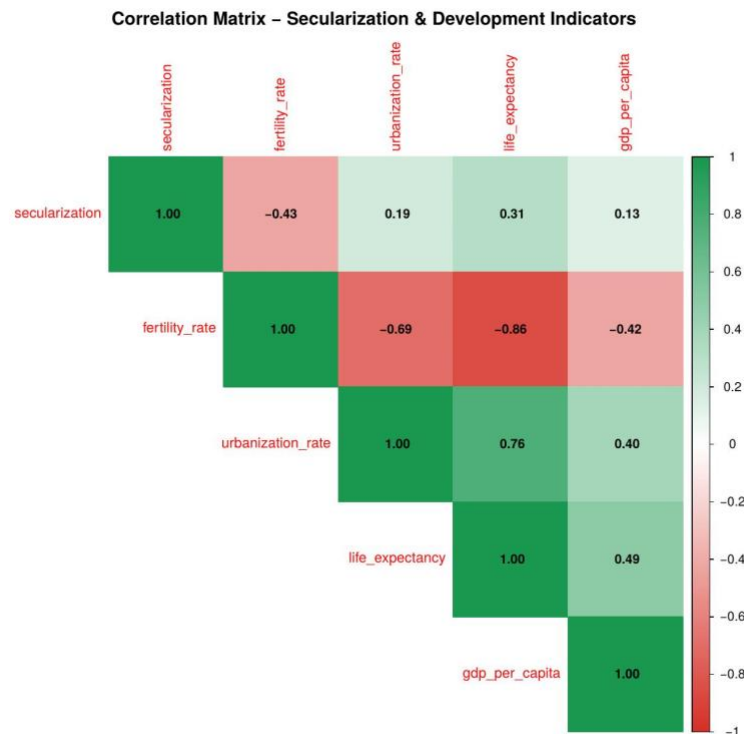


Figure 2. Correlation matrix of secularization and development indicators.

Secularization correlates negatively with fertility rate ($r = -0.43$, the strongest relationship it has with any predictor) and positively, but more weakly, with life expectancy (0.31) and urbanization (0.19). Its correlation with GDP per capita is the weakest of all (0.13). Just as important: the four predictors are themselves strongly related to each other — fertility correlates at -0.86 with life expectancy and -0.69 with urbanization. Notably, countries with fewer children per woman also tend to be more urban, live longer, and be richer, all at the same time. This overlap (called multicollinearity) makes it hard for a linear model to credit any one of these four factors individually, which is exactly the problem Sections 6 and onward address with several different tools.

3.3 Secularization vs. Each Predictor

To see the relationships directly rather than as a single number, we plot secularization against each predictor with a fitted straight line.

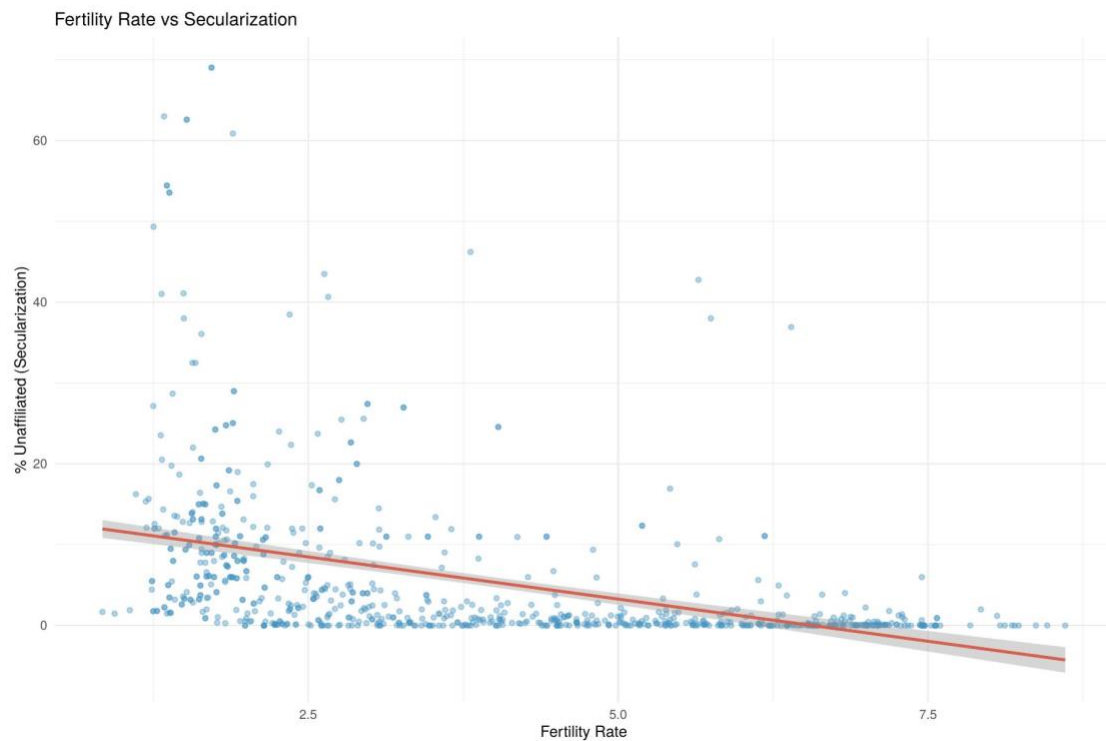


Figure 3. Fertility rate vs. secularization.

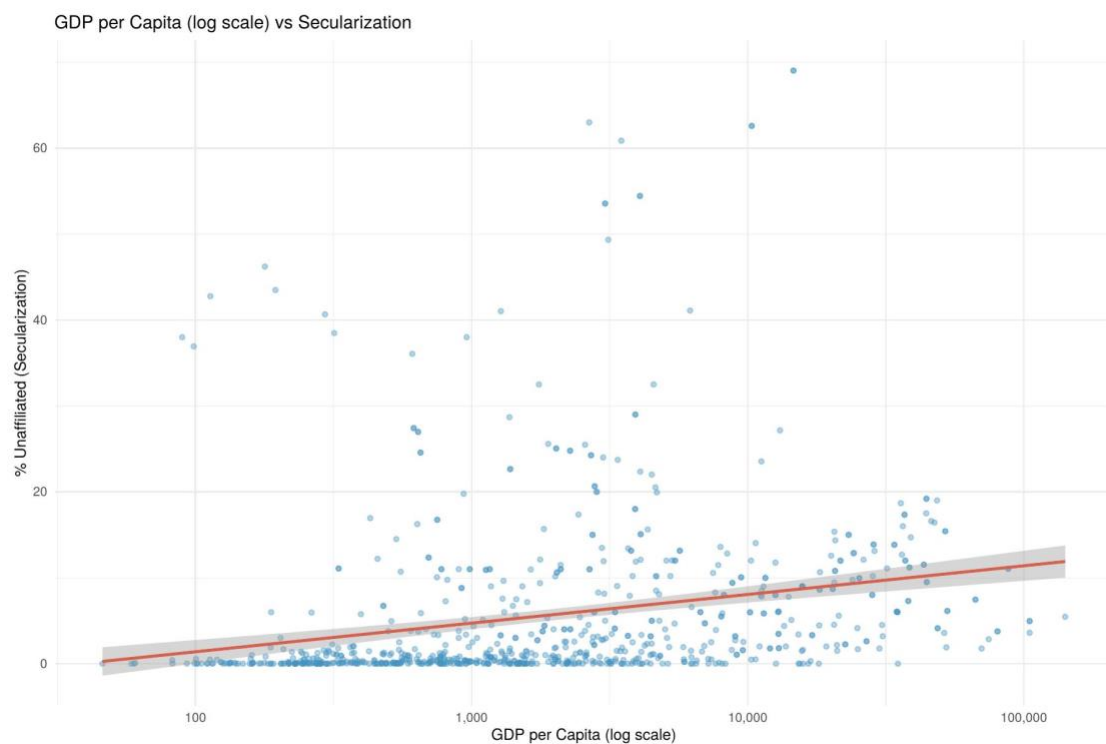


Figure 4. GDP per capita (log scale) vs. secularization.

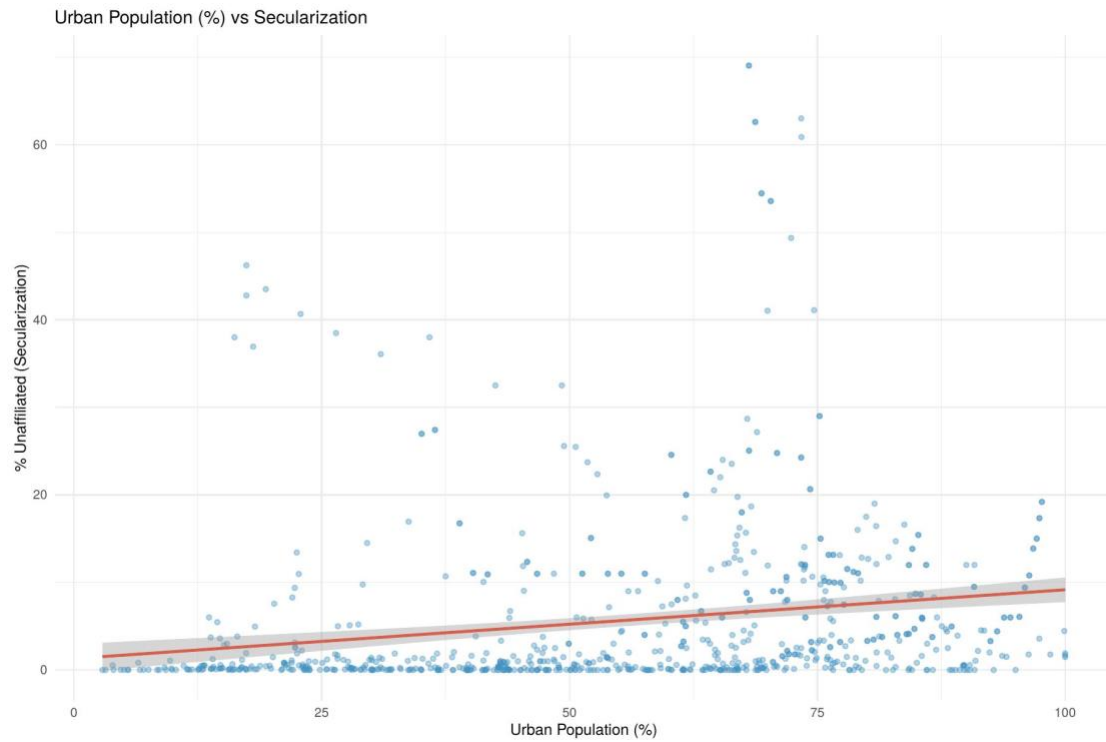


Figure 5. Urban population share vs. secularization.

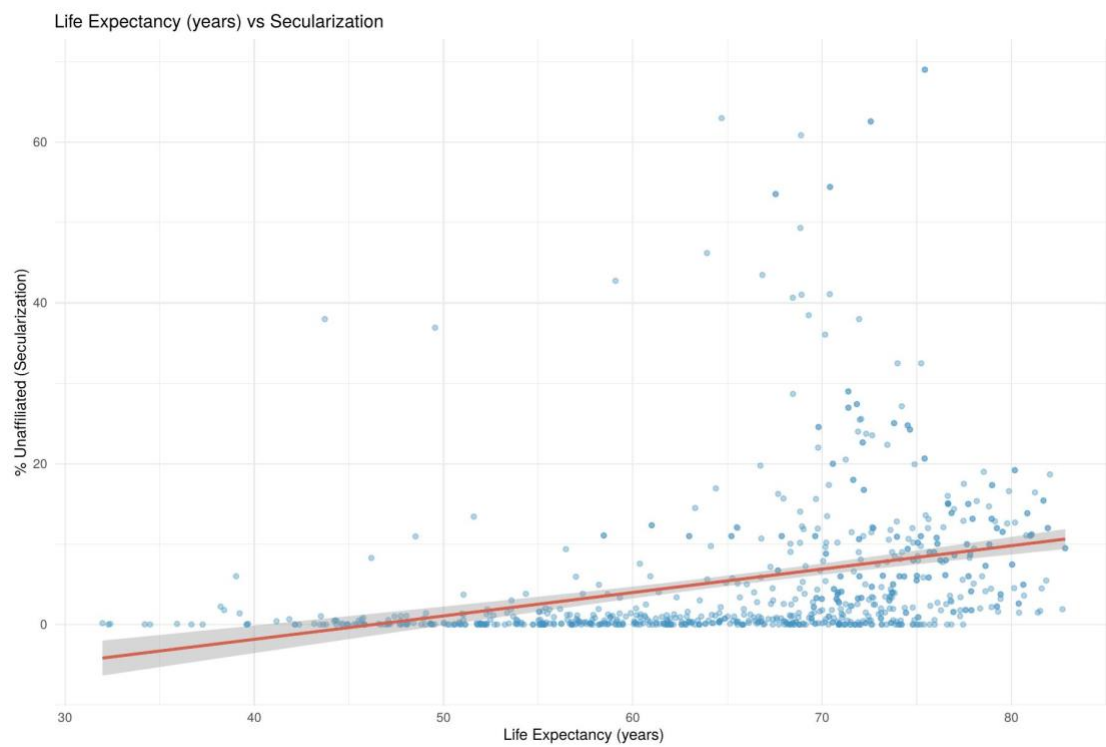


Figure 6. Life expectancy vs. secularization.

The fertility plot (Figure 3) shows the clearest pattern: almost all high-secularization countries have low fertility, although low fertility alone does not guarantee high secularization (the cloud of points is still wide at the low-fertility end). The GDP plot (Figure 4, log scale on the x-axis) shows a weak, scattered positive trend — many high-income countries are not particularly secular, and some lower-income countries are. Urbanization (Figure 5) and life expectancy (Figure 6) sit in between: a mild upward trend with a lot of noise around it.

3.4 Principal Component Analysis (PCA)

PCA looks at four variables together — secularization, fertility, urbanization, and GDP per capita, all converted to a common z-score scale — and asks: can we summarize most of what's going on with fewer than four numbers? It does this by finding new combined variables (“components”) that capture as much of the spread in the data as possible.

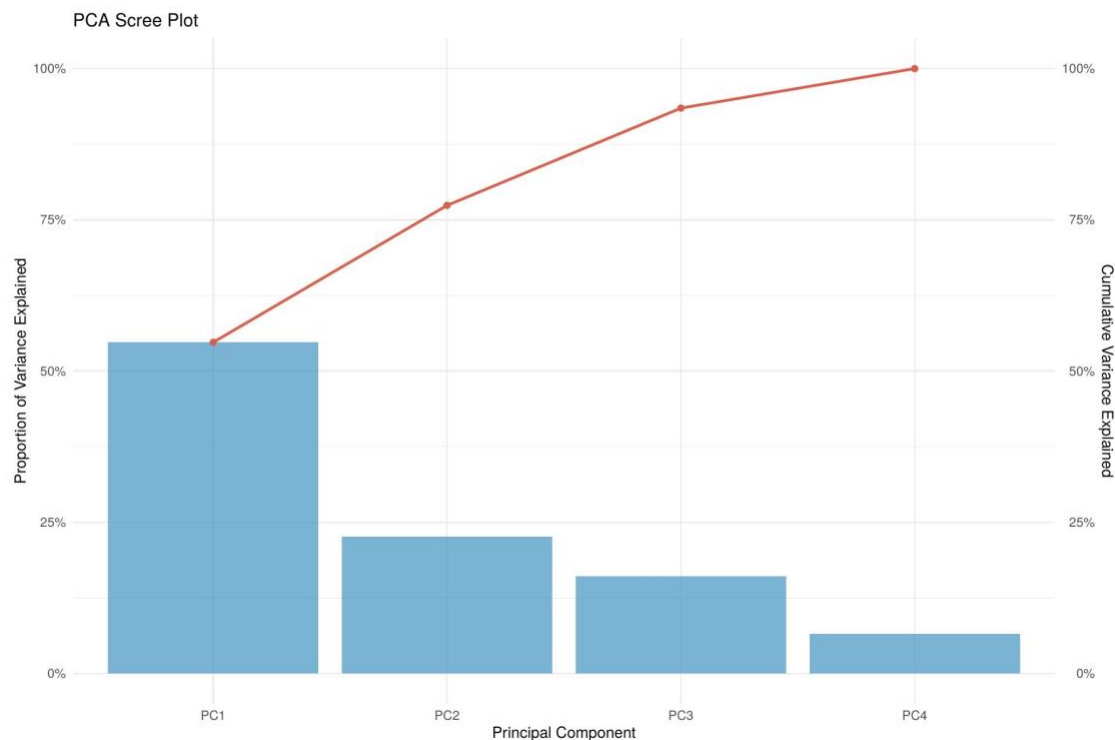


Figure 7. PCA scree plot: variance explained by each component.

The first component (PC1) alone captures 54.8% of the total variation in these four variables, and the first two components together capture 77.4%. In plain terms, this means the four variables are not telling four separate stories — they are largely telling one or two big stories (most likely, a general “modernization” story, plus a second story about how secularization sometimes breaks away from the income/urbanization pattern).

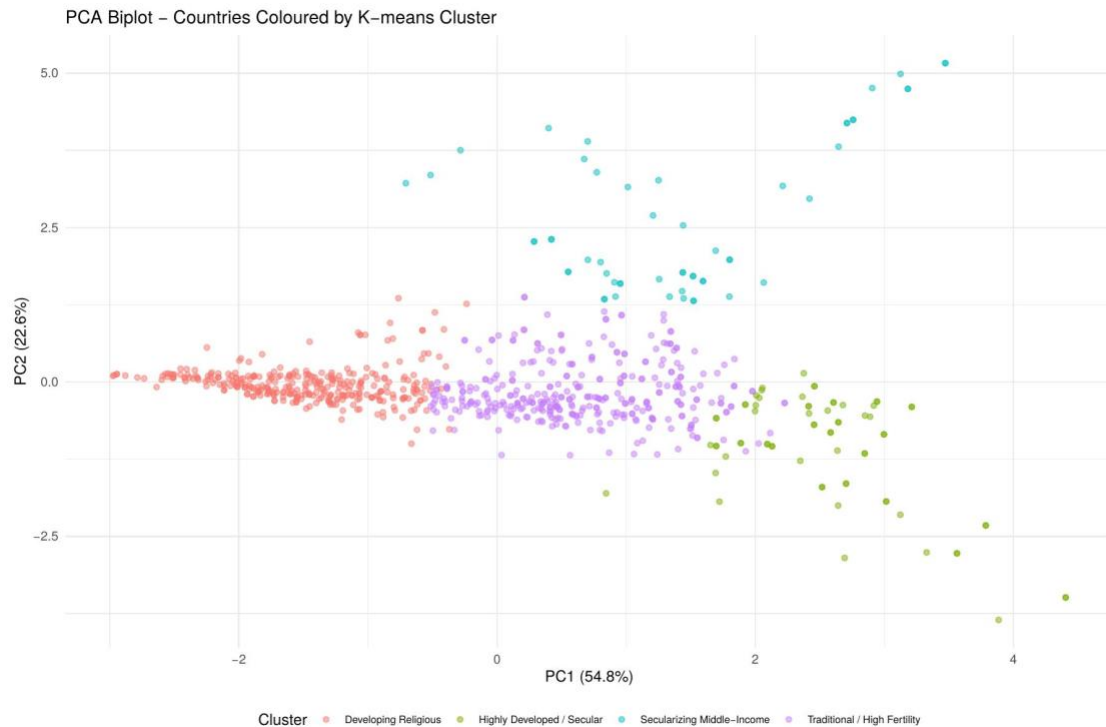


Figure 8. PCA biplot: countries plotted by PC1 and PC2, coloured by k-means cluster.

Plotting every country-year on just these two components (Figure 8) and colouring by the cluster found in Section 5 shows the four clusters occupying clearly separate regions of the plot. This is a good sign: it means the clustering in Section 5 is picking up on real structure in the data, not drawing arbitrary lines through a shapeless blob of points.

3.5 K-Means Clustering

K-means is an algorithm that sorts observations into a chosen number of groups (k) by repeatedly assigning each point to its nearest group centre and then recalculating the centre, until the groups stop changing. Because it measures “nearness” using ordinary distance, all four clustering variables (secularization, fertility, urbanization, GDP per capita) are first rescaled to a common z-score scale — otherwise GDP, which ranges into the tens of thousands, would dominate the distance calculation and the other three variables would barely matter.

3.5.1 Choosing the Number of Clusters

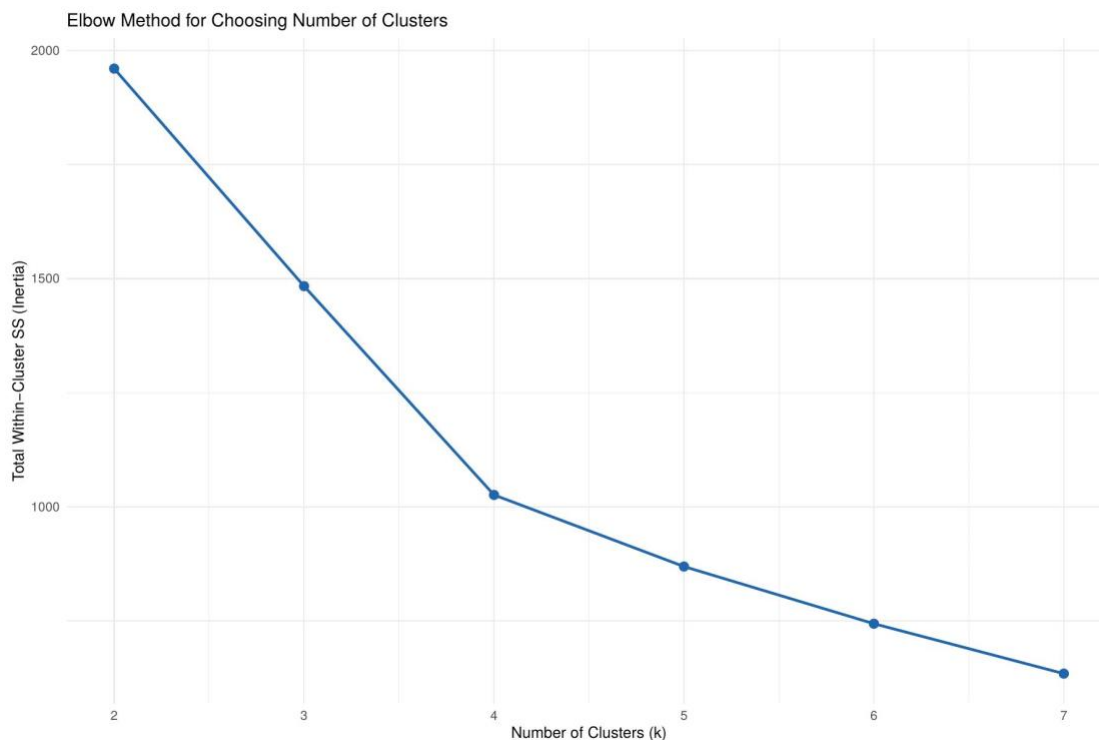


Figure 9. Elbow method: total within-cluster variation vs. number of clusters.

To choose how many groups (k) to use, we try $k = 2$ through 7 and plot how much “within-group spread” remains for each choice. There is a clear bend (“elbow”) at $k = 4$: going from 3 to 4 groups reduces the spread substantially, but going beyond 4 groups gives much smaller improvements. We therefore use $k = 4$.

3.5.2 Cluster Profiles

Running k-means with $k = 4$ produces four groups with these average characteristics:

Cluster label	n	Secular. (%)	Fertility	Urban. (%)	GDP/cap (USD)
Developing Religious	329	1.09	5.88	33.3	880
Highly Developed / Secular	70	9.64	1.77	81.0	45,834
Secularizing Middle-Income	56	34.50	2.29	56.7	3,306
Traditional / High Fertility	356	4.52	2.65	70.7	5,659

This shows that the two large groups (Developing Religious and Traditional/High Fertility, together 685 of 811 clustered observations) have low secularization and represent the bulk of the world's country-years in this sample. The interesting exception is the small Secularizing Middle-Income group: it does not have the highest income or the lowest fertility of the four groups, yet it has by far the highest secularization (34.5%, almost four times the Highly Developed/Secular group's 9.6%). This single comparison is the clearest piece of evidence in this report that secularization is not just a side-effect of being rich.

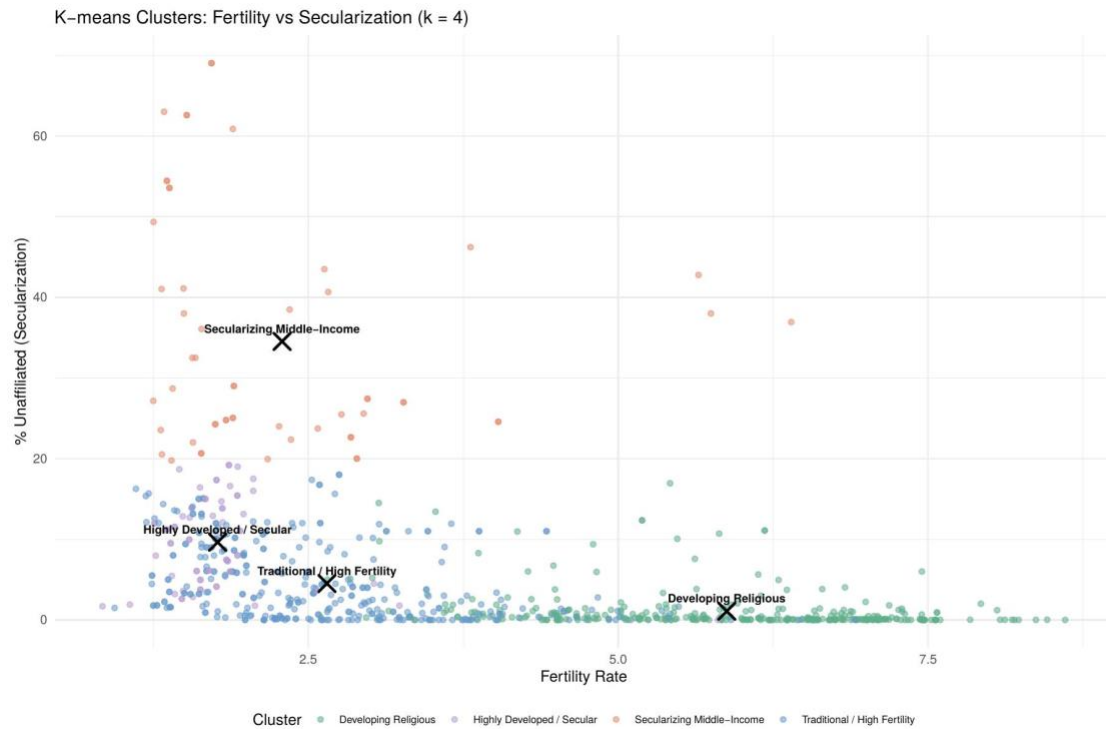


Figure 10. K-means clusters: fertility rate vs. secularization, with cluster centres (X) and labels.

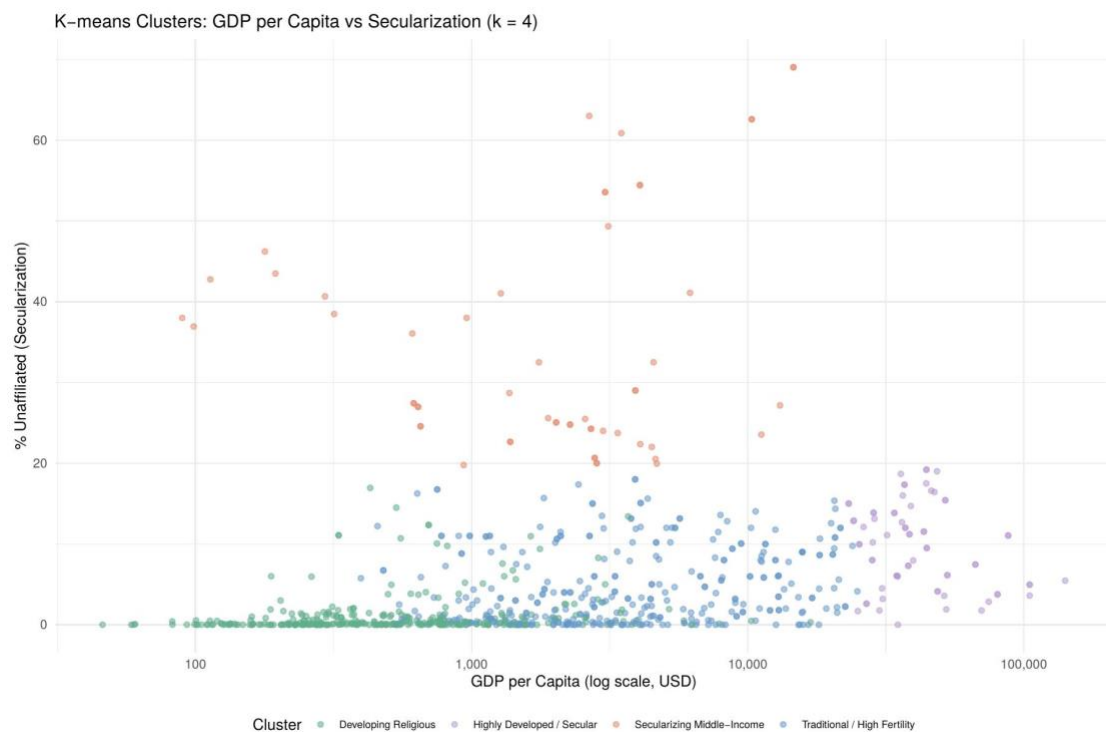


Figure 11. K-means clusters: GDP per capita (log scale) vs. secularization.

Figure 10 shows the Secularizing Middle-Income group (orange points) sitting noticeably higher on the secularization axis than any other group, despite overlapping with the other groups on fertility. Figure 11 makes the income point even clearer: this group's GDP values sit in the low-to-middle range of the x-axis, well below the Highly Developed/Secular group's, yet its secularization values are still the highest in the plot.

3.6 Hierarchical Clustering (a Second Opinion)

K-means requires choosing k in advance and can land on a different answer depending on its starting point. As a check, we also ran hierarchical clustering, a different algorithm that builds a tree of nested groups from the bottom up by repeatedly merging the two closest points or groups, without needing to pick k beforehand.

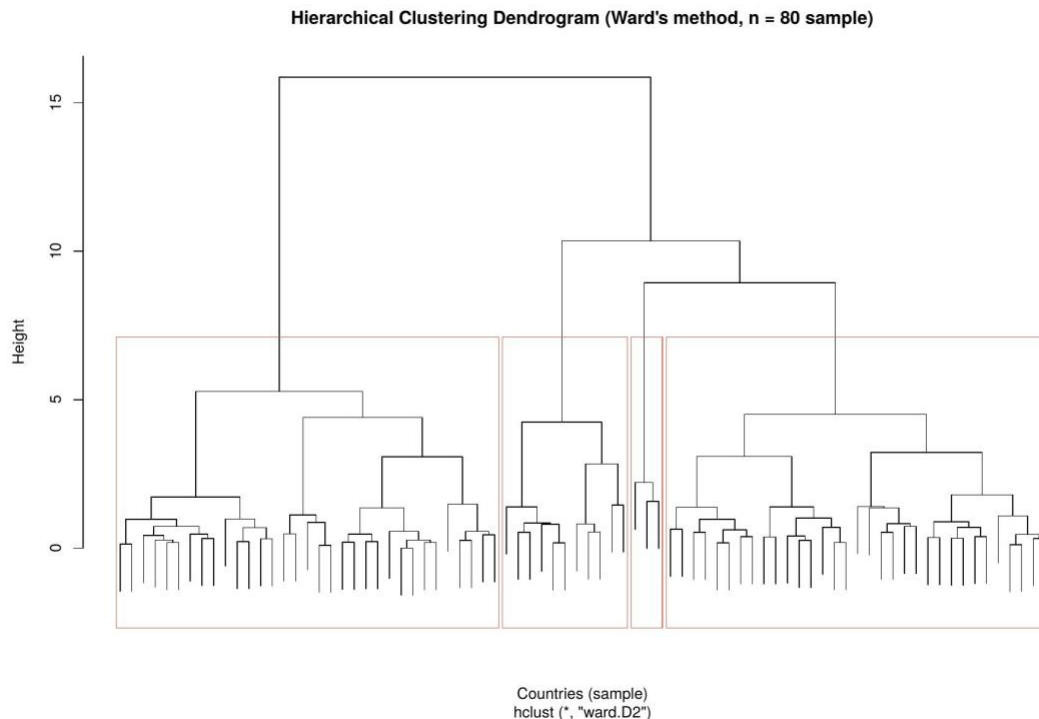


Figure 12. Hierarchical clustering dendrogram (Ward's method, 80-country sample).

Cutting this tree into four groups and comparing them to the k-means labels gives the following agreement table:

K-means label	Hier. 1	Hier. 2	Hier. 3	Hier. 4
Developing Religious	325	0	4	0
Highly Developed / Secular	0	0	4	66
Secularizing Middle-Income	0	28	28	0
Traditional / High Fertility	51	0	283	22

Reading this table: hierarchical group 1 almost perfectly matches Developing Religious (325 of 329 countries), and hierarchical group 4 almost perfectly matches Highly Developed/Secular (66 of 70). Traditional/High Fertility splits mainly into hierarchical group 3, but with some overlap elsewhere, and Secularizing Middle-Income splits roughly evenly between hierarchical groups 2 and 3 rather than forming one clean matching group. In plain terms: two completely different clustering algorithms, asked the same question in different ways, strongly agree on the two

largest, most distinct groups (Developing Religious and Highly Developed/Secular), and partially agree on the other two, smaller groups. This gives confidence in the two biggest clusters of the k-means solution while suggesting the boundary between Traditional/High Fertility and Secularizing Middle-Income is somewhat less sharp — consistent with k-means having to draw one specific dividing line through what may be a more gradual transition in the underlying data.

PART II — SUPERVISED LEARNING

4. Regression Analysis

4.1 Simple OLS: Fertility Rate Alone

As a baseline, we fit the simplest possible model: predict secularization using fertility rate alone, with ordinary least squares (OLS) — the standard “best-fit straight line” method.

$$\text{secularization} = 16.16 - 2.37 \times \text{fertility_rate}$$

Both numbers in this equation are highly statistically significant ($p < 0.001$), meaning we can be confident this is a real pattern and not random noise. In plain terms: each additional child per woman is associated with about a 2.4 percentage-point drop in the share of the population that is unaffiliated. However, this simple model only explains 15.4% of the variation in secularization ($R^2 = 0.154$) — useful, but far from the whole story, which is expected since we are deliberately ignoring three other relevant factors at this stage.

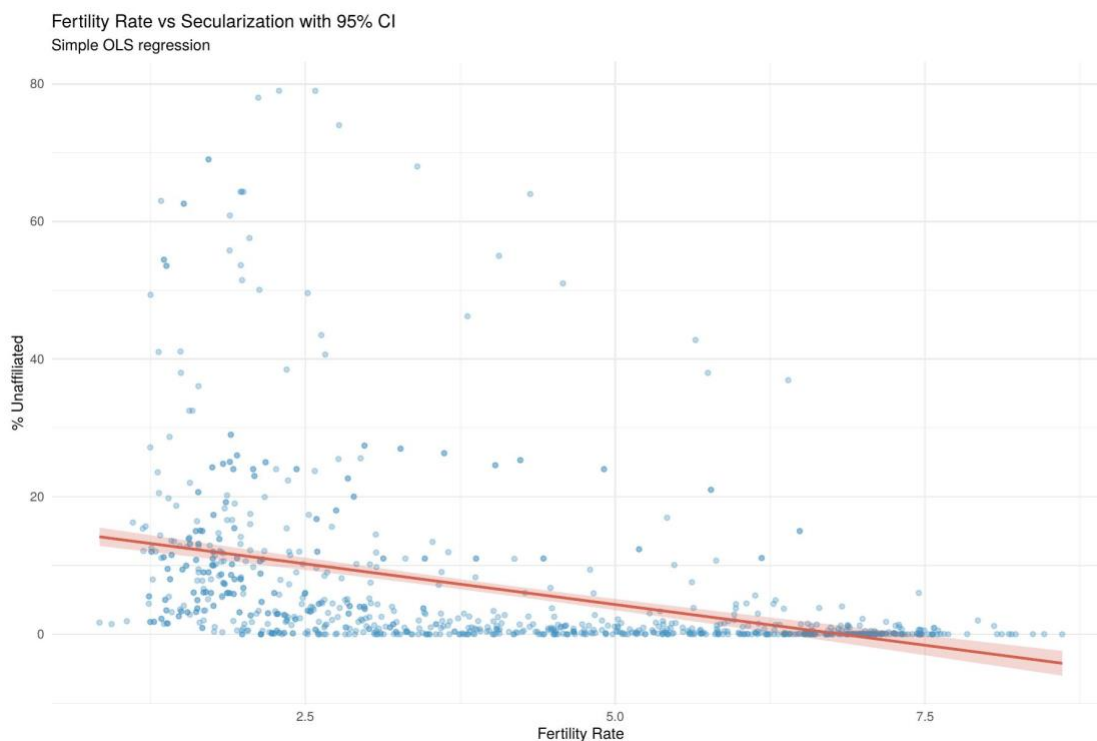


Figure 13. Simple OLS fit of secularization on fertility rate, with 95% confidence band.

4.2 Multivariate OLS: All Four Predictors Together

Adding urbanization, life expectancy, and GDP per capita into the same model ($n = 808$, smaller than before because some countries are missing GDP data in some years) gives:

Predictor	Estimate	Std. Error	p-value	Significant?
(Intercept)	26.77	4.91	<0.001	Yes
<i>fertility_rate</i>	-3.057	0.300	<0.001	Yes
<i>urbanization_rate</i>	-0.067	0.020	0.001	Yes (weakly)
<i>life_expectancy</i>	-0.084	0.066	0.201	No
<i>gdp_per_capita</i>	-0.0000146	0.0000247	0.554	No

Adding the three extra predictors improves the model only modestly, to $R^2 = 0.205$ (up from 0.154). Fertility remains the clear standout: it is the only predictor with both a large effect and strong statistical significance. Urbanization is statistically significant but its effect size is tiny. Life expectancy and GDP per capita are not statistically distinguishable from zero once the other variables are included — in plain terms, once we know a country's fertility rate, knowing its income or life expectancy adds almost nothing extra for predicting secularization.

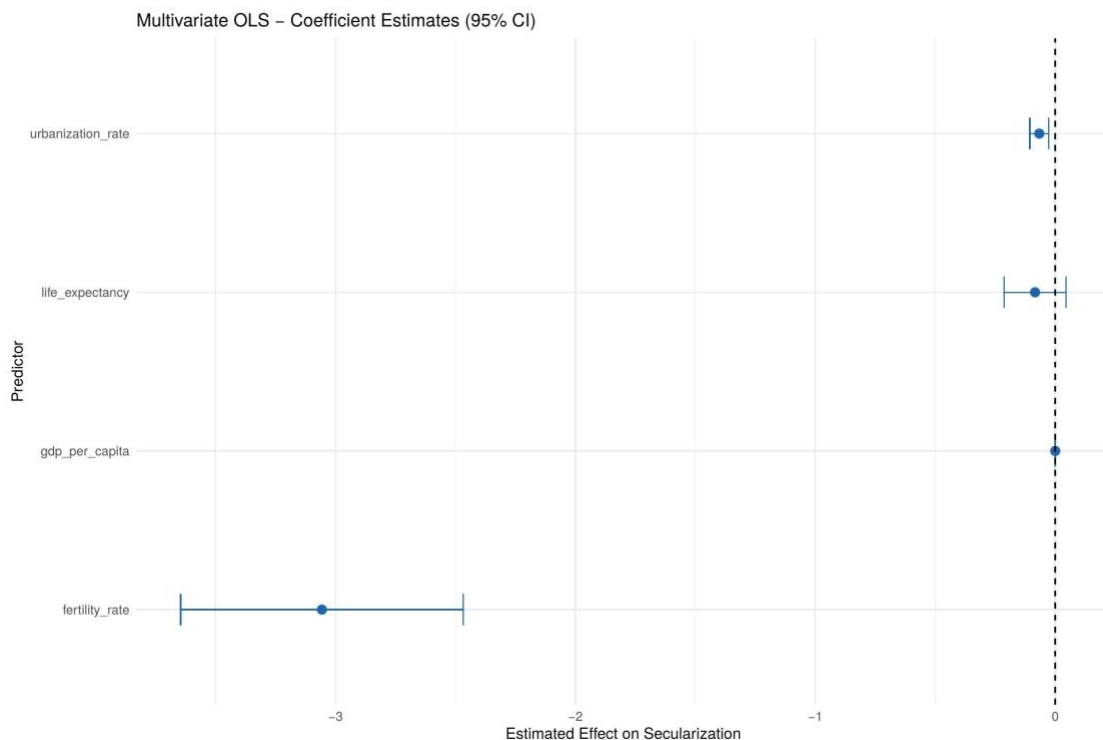


Figure 14. Multivariate OLS coefficient estimates with 95% confidence intervals.

In Figure 14, fertility's confidence interval (the horizontal line through its dot) is the only one clearly separated from zero. This is a visual way of seeing the same conclusion as the table above.

4.2.1 Checking the Model's Assumptions (Residual Diagnostics)

Before trusting a linear regression's numbers, it's good practice to check whether the model's mistakes (residuals) behave the way OLS assumes — roughly evenly scattered, with no obvious pattern, and roughly bell-shaped.

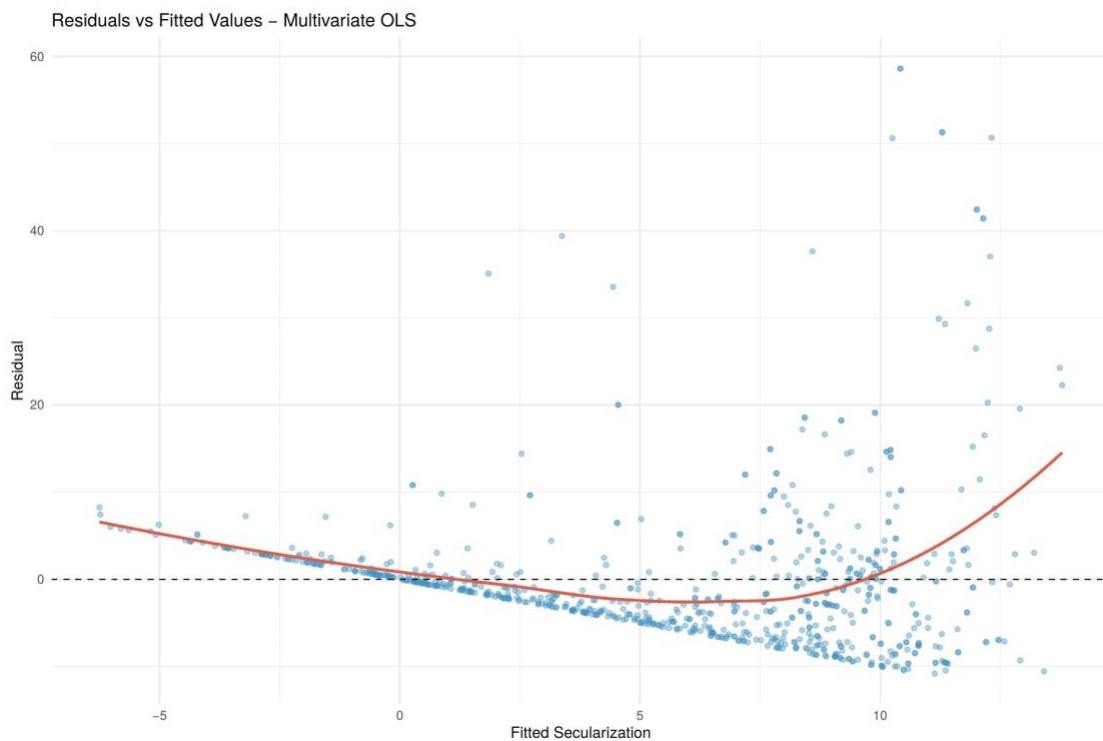


Figure 15. Residuals vs. fitted values for the multivariate OLS model.

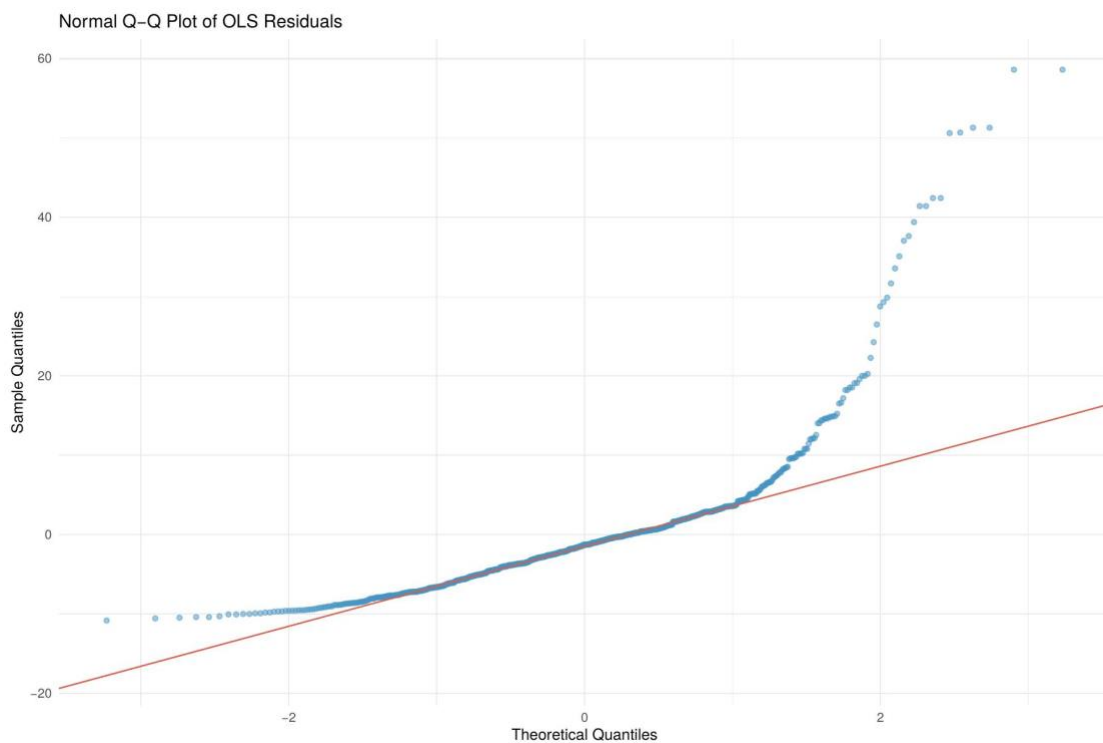


Figure 16. Normal Q-Q plot of the OLS residuals.

Figure 15 plots each prediction error against the model's predicted value; in a well-behaved model, this cloud of points should look like a flat, even band centred on zero with no funnel shape. Here, the spread visibly widens for higher fitted values and a cluster of large positive errors appears at low fitted values, indicating the model's errors are not constant (heteroscedasticity) and that it systematically under-predicts a subset of highly secular countries. The Q-Q plot (Figure 16) compares the residuals' distribution to a perfect bell curve; the points should hug the diagonal line if residuals are normally distributed. Here, the upper tail clearly bends away from the line, confirming the residuals are right-skewed — again pointing to the same high-secularization countries the model struggles with elsewhere in this report. In simple terms, this is more evidence that a straight-line model is not fully appropriate for this relationship, motivating the non-linear Random Forest model in Section 5.

4.2.2 Does Fertility's Effect Depend on Urbanization?

A natural question is whether **fertility matters equally in rural and highly urbanized countries**, or whether urbanization changes how strongly fertility predicts secularization. To investigate this, an **interaction term (fertility × urbanization)** was added to the multivariate model.

The results provide **little evidence that urbanization modifies the fertility–secularization relationship**. The interaction coefficient is **small and statistically insignificant (coefficient = -0.0092 , $p = 0.244$)**, indicating that any interaction effect is weak at best.

Model performance also changes only negligibly. The **adjusted R^2 increases from 0.2009 to just 0.2012**, a difference that is too small to be substantively meaningful.

The key conclusion is that fertility's effect on secularization appears remarkably stable across different levels of urbanization. Whether a country is relatively rural or highly urbanized, fertility remains a strong predictor of secularization. This serves as an important **robustness check**, reinforcing the central finding of this report that **fertility is the dominant predictor of secularization**.

4.3 Ridge and Lasso Regression

Section 3.2 showed that the four predictors strongly overlap with each other (multicollinearity), which can make ordinary OLS coefficients unstable. Ridge and Lasso are two regression methods designed specifically for this situation: both add a “penalty” that discourages large coefficients, but Lasso's penalty can shrink unhelpful coefficients all the way to exactly zero (effectively removing them from the model), while Ridge only shrinks them smaller without removing them. The right amount of penalty (λ , “lambda”) for each method is chosen automatically using cross-validation, so we don't have to guess it by hand.

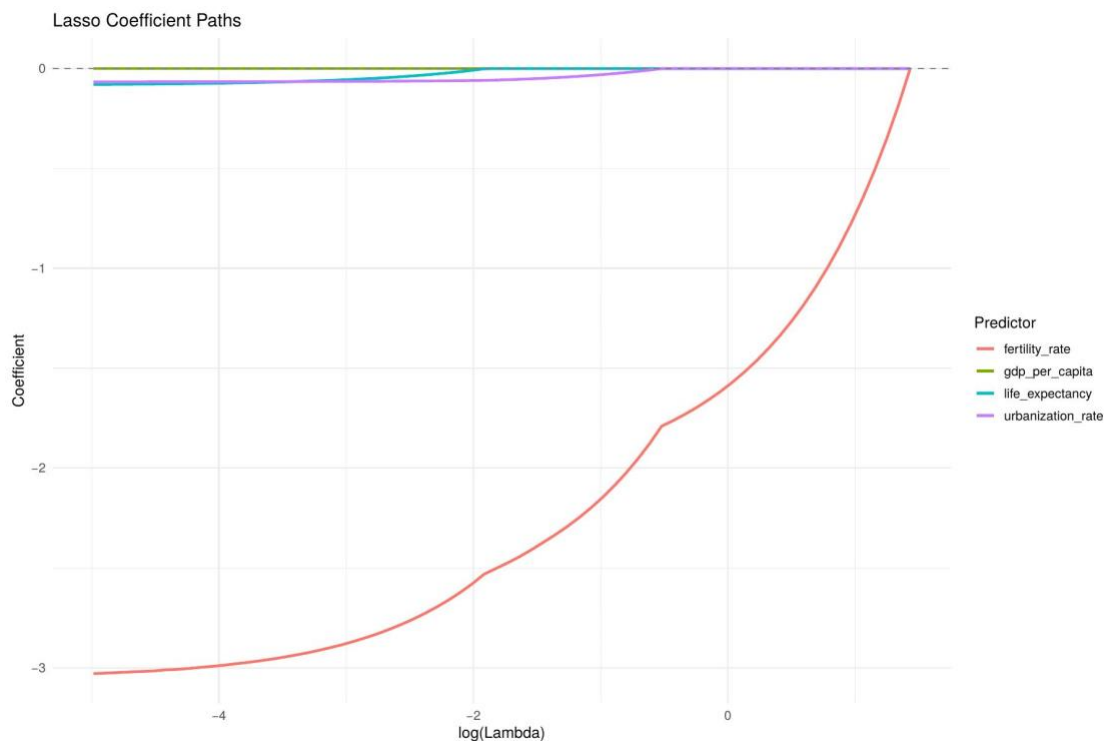


Figure 17. Lasso coefficient paths as the penalty strength (λ) increases.

Figure 17 shows what happens to each predictor's coefficient as the Lasso penalty grows from left (no penalty, behaves like OLS) to right (very strong penalty, shrinks everything to zero). The key pattern: *fertility_rate*'s line stays furthest from zero for the longest — it is the last coefficient Lasso is willing to shrink away — while *gdp_per_capita* and *life_expectancy* collapse toward zero much earlier. This is an independent confirmation, from a completely different method than OLS, that fertility is the predictor doing most of the real work.

At each method's own best (cross-validated) penalty strength, both Ridge and Lasso keep fertility's coefficient large while shrinking the other three. The table below compares all three methods directly:

Predictor	OLS	Ridge ($\lambda=0.419$)	Lasso ($\lambda=0.0068$)
<i>fertility_rate</i>	-3.057	-2.606	-3.029
<i>urbanization_rate</i>	-0.0666	-0.0581	-0.0664
<i>life_expectancy</i>	-0.0841	-0.0239	-0.0794
<i>gdp_per_capita</i>	-0.0000146	-0.0000148	-0.0000143

Notice that fertility's coefficient barely shrinks under either penalty (from -3.057 to -2.606 under Ridge, and to -3.029 under Lasso, which keeps it almost unchanged), while life expectancy's coefficient shrinks much more under Ridge (from -0.084 to -0.024, roughly a 70% reduction). Neither method actually sets any coefficient to exactly zero at its own best penalty strength, which is itself informative: it suggests all four predictors carry at least some independent signal, but fertility's signal is by far the strongest and most resistant to being shrunk away. Ridge and Lasso's cross-validated prediction error (mean-squared error of 77.76 and 77.63 respectively) is, in this case, very close to plain OLS's in-sample error (76.76) — regularization here mainly stabilizes the coefficients

rather than dramatically improving prediction accuracy, which makes sense given only four, only moderately correlated predictors.

5. Random Forest Regression (Tree-Based Methods)

To address the non-linearity hinted at throughout Part I and Section 4 (recall the residual diagnostics in 4.2.1), we fit a Random Forest: an ensemble method that builds many individual decision trees, each on a slightly different random sample of the data and a random subset of predictors, and then averages their predictions. This averaging usually makes Random Forest much better than a single tree (or a straight line) at capturing bends and interactions in the data, at the cost of being harder to summarize with a single equation.

5.1 Model Performance

The model was first trained on a 70% random sample of the data and evaluated on the remaining, unseen 30% (n = 939 total, complete cases):

Model	R ²	RMSE
Simple OLS (fertility only)	0.154	11.36
Multivariate OLS	0.205	8.79
Random Forest (single test split)	0.561	6.04

Random Forest more than doubles the explained variance of the best linear model. But a single 70/30 split could be lucky or unlucky depending on which countries happen to land in the test set — so we don't stop here.

5.2 Checking Robustness with Cross-Validation

To make sure the strong Random Forest result above isn't a fluke of one particular split, we repeat the same training-and-testing process five times, each time leaving out a different 20% of the data as the test set (5-fold cross-validation), and average the results.

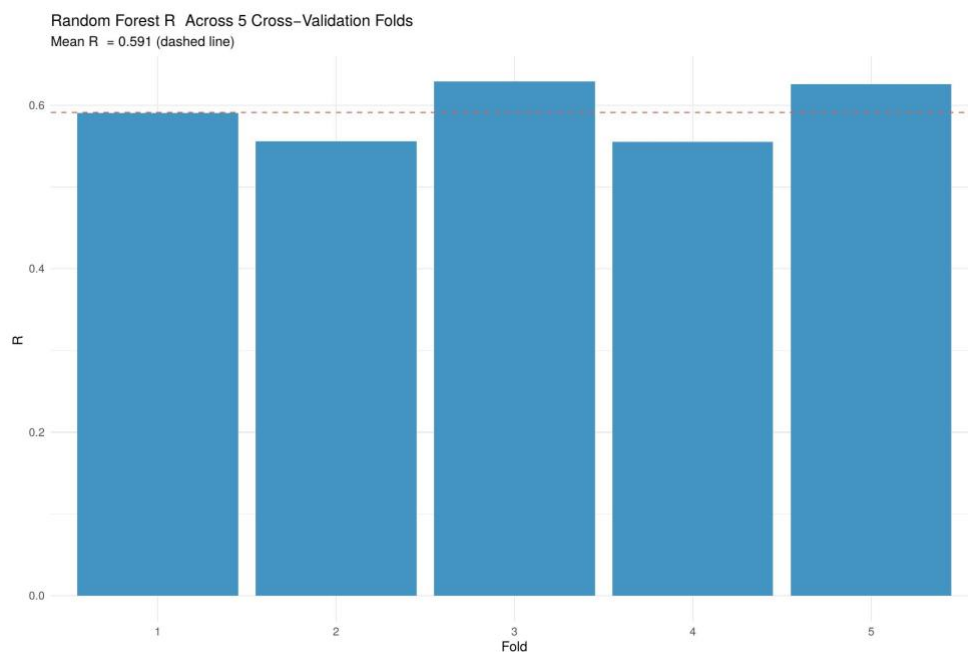


Figure 18. Random Forest R^2 across 5 cross-validation folds.

The R^2 across the five folds ranges narrowly, with a mean of 0.591 — actually slightly higher than, and consistent with, the single-split result of 0.561 from Section 5.1. In plain terms: the Random Forest's strong performance is a stable, repeatable property of this dataset, not a one-off lucky split. This is exactly the kind of check the simple train/test split in Section 5.1 could not provide on its own.

5.3 Which Predictors Matter Most?

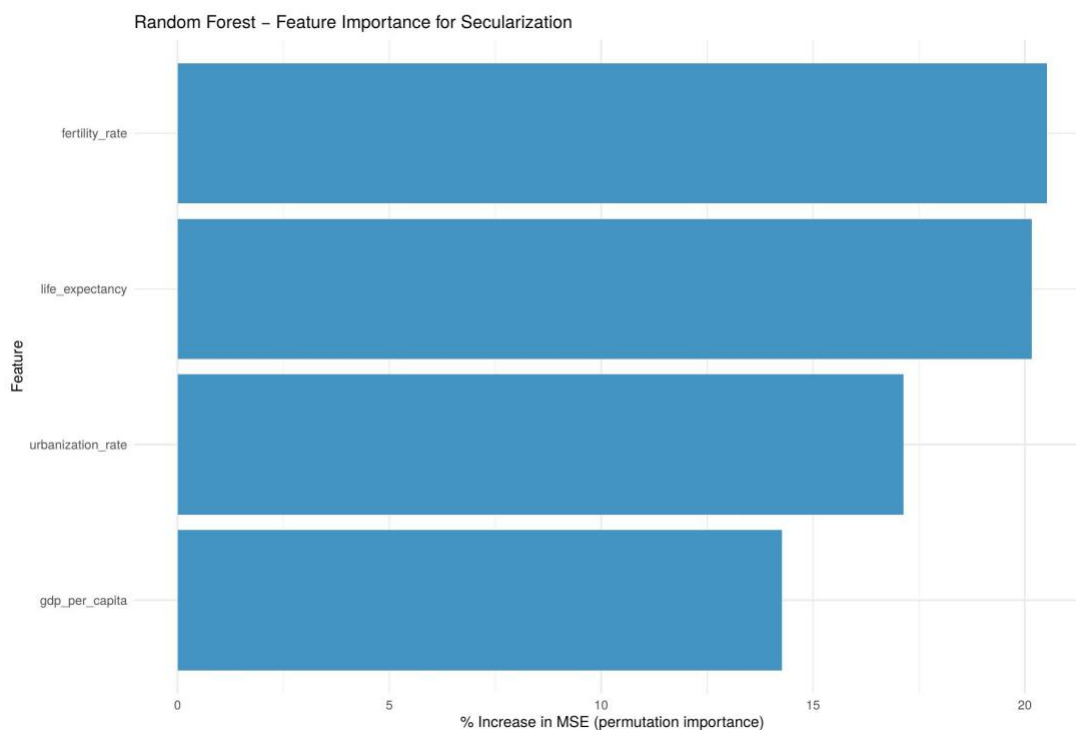


Figure 19. Random Forest permutation feature importance.

Permutation importance works by randomly scrambling one predictor at a time and measuring how much worse the model's predictions get; a bigger drop in accuracy means the model relied on that predictor more heavily. Fertility rate's importance score is roughly 4–5 times larger than the next-most-important predictor (life expectancy), with urbanization and GDP per capita contributing comparatively little.

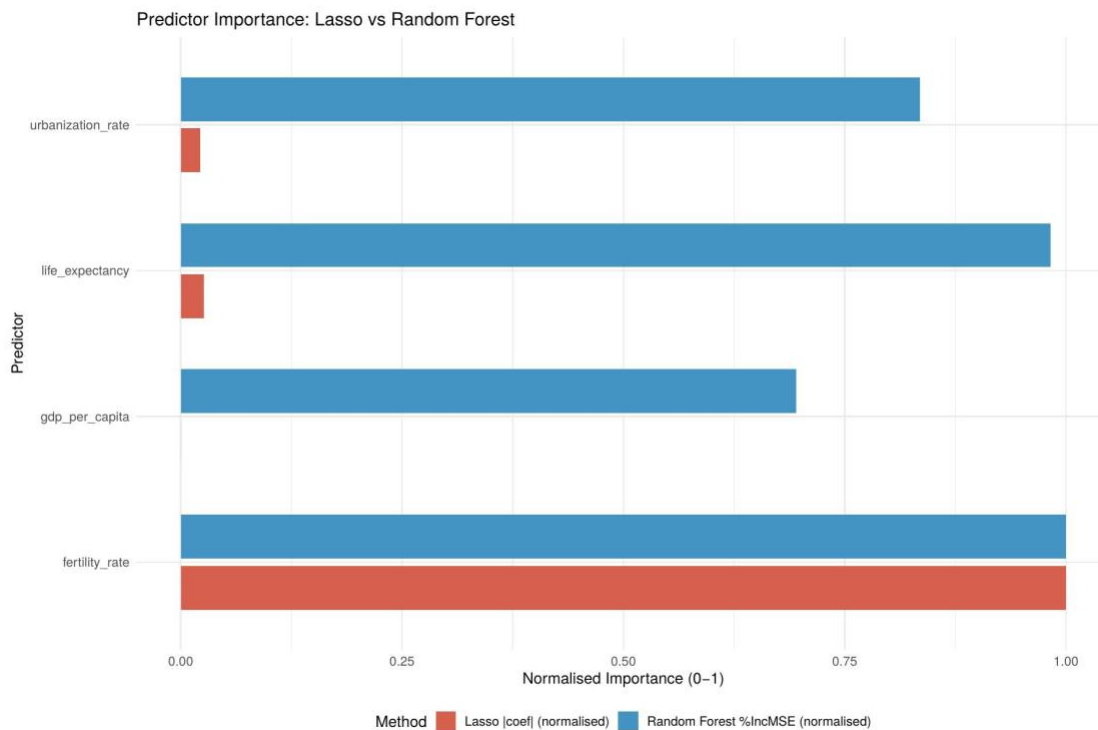


Figure 20. Predictor importance: Lasso vs. Random Forest, side by side.

Figure 20 puts the Lasso and Random Forest importance rankings next to each other on the same 0-to-1 scale. Both a linear, penalized method and a non-linear, ensemble method — built on very different mathematical principles — agree that fertility rate is the dominant predictor, while GDP per capita ranks lowest (or close to lowest) in both. Agreement between two very different methods is much stronger evidence than either method's ranking on its own.

5.4 Actual vs. Predicted Values

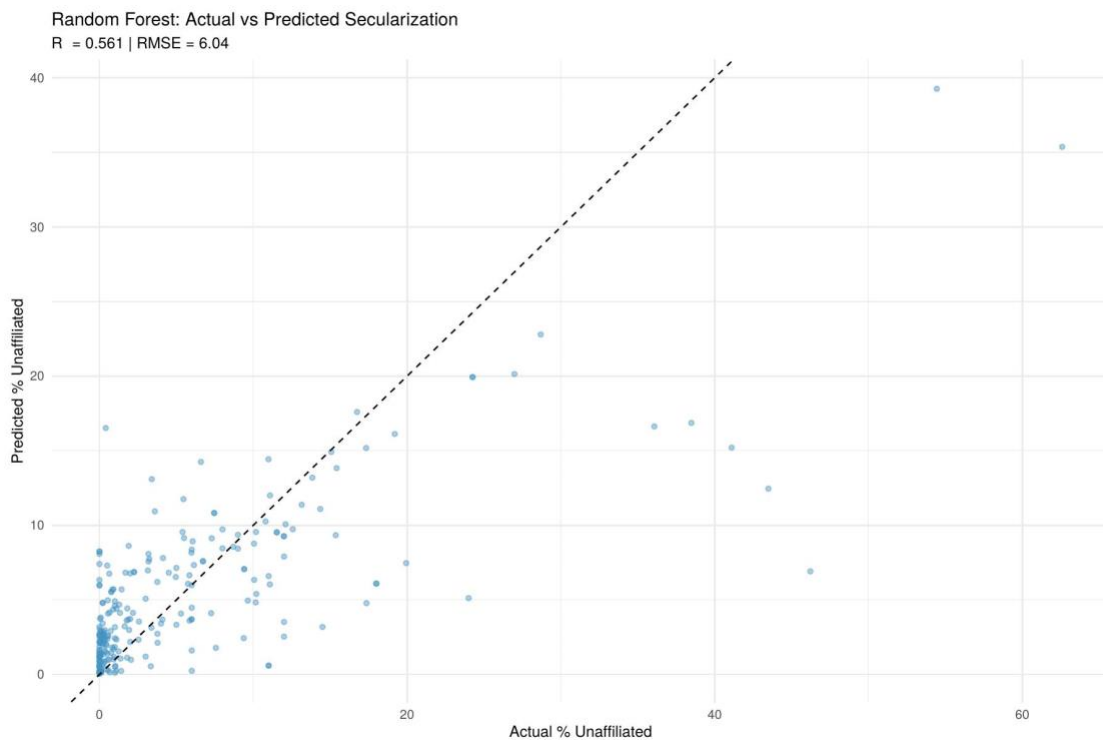


Figure 21. Random Forest: actual vs. predicted secularization on the test set.

Each point in Figure 21 compares one country-year's true secularization value (x-axis) to the model's prediction (y-axis); perfect predictions would fall exactly on the dashed diagonal line. Most points cluster near the diagonal for countries with low-to-moderate secularization (below about 20%), showing the model does well for the bulk of the data. For the handful of countries above 40% secularization, however, the model's predictions fall noticeably below the diagonal — it systematically under-predicts these extreme cases. This is a common limitation when a target variable is skewed (most countries have low secularization, very few have very high secularization) and there are correspondingly few training examples at the extreme end for the model to learn from.

5.5 How Does Fertility Actually Affect Secularization?

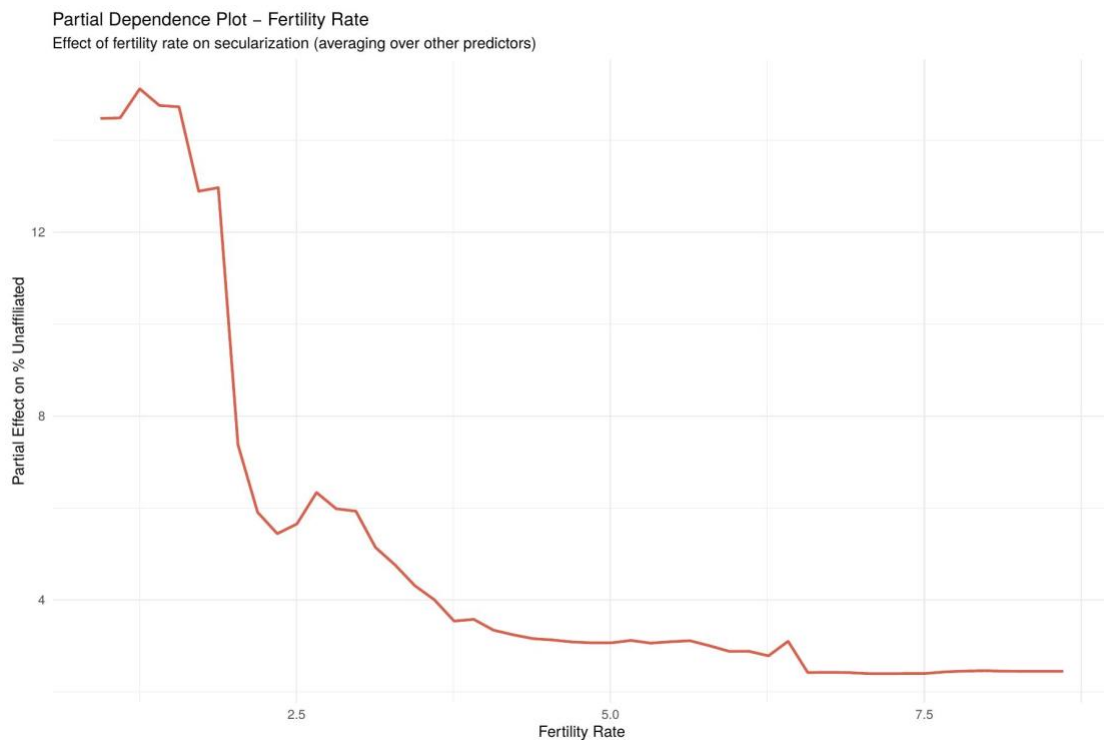


Figure 22. Partial dependence plot: the effect of fertility rate on predicted secularization.

A partial dependence plot answers a focused question: holding the other three predictors at their typical values, what does the model predict secularization to be at each possible fertility level? The curve in Figure 22 rises steeply as fertility falls from about 7.5 to 5 children per woman, then flattens out (and even dips slightly) at lower fertility levels. A straight line, by construction, cannot bend this way — it would have to be one constant slope all the way across the graph. This single plot is the clearest visual explanation in the whole report for why Random Forest's R^2 (0.561–0.591) so clearly beats both OLS models' R^2 (0.154 and 0.205): the true relationship has a bend in it, and only the non-linear model can follow that bend.

6. Conclusions

This report set out to identify which socio-economic and demographic factors best predict secularization, and to compare what different statistical learning methods reveal about that relationship using the same underlying data.

The results are remarkably consistent across every method used. **Fertility rate emerges as the single most important and most consistent predictor of secularization.** It has the strongest raw correlation, remains the dominant predictor in OLS, Ridge, and Lasso models, and is ranked as the most important feature by Random Forest. **The relationship is also non-linear rather than strictly linear**, with secularization rising sharply as fertility falls and then flattening at lower fertility levels. As a result, **the Random Forest model substantially outperforms linear regression ($R^2 \approx 0.56\text{--}0.59$, confirmed through 5-fold cross-validation) compared with $R^2 = 0.205$ for the multivariate OLS model.**

Income, by contrast, appears to be a weak and unreliable predictor of secularization. Its raw correlation with secularization is small, it is not statistically significant in the multivariate OLS model, and it is among the first coefficients that Lasso shrinks toward zero. The clustering analysis reinforces this conclusion. **The most secular group of countries is not the richest group.** The **Secularizing Middle-Income** cluster exhibits the highest average secularization (**34.5%**), while the wealthiest cluster (**Highly Developed / Secular**) has a substantially lower average secularization (**9.6%**). Importantly, **both k-means and hierarchical clustering identify the same four-group structure**, while PCA reveals these groups as clearly distinct regions in a two-dimensional representation of the data.

Taken together, the evidence points to a more nuanced explanation than the simple claim that “richer countries are automatically more secular.” Instead, **demographic change—captured here primarily through declining fertility—appears to be the most universal and important correlate of secularization.** Urbanization, life expectancy, and income still contribute information, but their effects are smaller, overlapping, and individually weaker once fertility is taken into account.

Several limitations should be acknowledged. **The data are observational and therefore cannot establish causality.** It is entirely possible that secularization influences fertility through mechanisms such as delayed marriage or changing family formation patterns, rather than fertility affecting secularization alone. The merged dataset is also moderately sized (**954 observations**, falling to **808 observations** in the multivariate analysis because of missing GDP data) and is unevenly distributed across countries and years. Finally, although **the choice of four clusters is supported by both the elbow method and an independent clustering approach**, cluster selection inevitably involves some degree of researcher judgment.

Future research could extend this analysis using panel-data methods that exploit repeated observations of the same countries over time. Additional predictors such as education, institutional quality, or political regime type may also help explain variation in secularization beyond the demographic factors examined here.

References

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Appendix: R Code

The complete R script is submitted alongside this report as WRP_2020_analysis.R. Key sections are reproduced below for reference. It reads the three raw source files (WRP_national.csv, COW-country-codes.csv, WDIData.csv), cleans and merges them, then runs the full sequence of analyses described above: exploratory data analysis, OLS (simple, multivariate, with interaction, with residual diagnostics), Ridge/Lasso, Random Forest with cross-validation, k-means clustering, hierarchical clustering, and PCA.

A. Data Loading and Merging

```
# — 1. DATA LOADING —————

# Adjust these paths to match your local folder structure
wrp_path <- "/Users/srishtikansra/Documents/uni of milan/ML_Stats/WRP_2020/WRP_national.csv"
cow_path <- "/Users/srishtikansra/Documents/uni of milan/ML_Stats/WRP_2020/COW-country-codes.csv"
wdi_path <- "/Users/srishtikansra/Documents/uni of milan/ML_Stats/WRP_2020/wdi-csv-zip-57-mb/WDIData.csv"

cat("Loading WRP national dataset...\n")
wrp_raw <- read_csv(wrp_path, show_col_types = FALSE)

cat("Loading COW country codes...\n")
cow_raw <- read_csv(cow_path, show_col_types = FALSE)

cat("Loading World Bank WDI data (large file, this may take a moment)...\n")
wdi_raw <- read_csv(wdi_path, show_col_types = FALSE)
```

B. Multivariate OLS Regression

```
# — 9. MULTIVARIATE OLS REGRESSION —————

reg_df_multi <- df %>%
  select(secularization, fertility_rate, urbanization_rate,
  life_expectancy, gdp_per_capita) %>%
  drop_na()

model_multi <- lm(
  secularization ~ fertility_rate + urbanization_rate +
  life_expectancy + gdp_per_capita,
  data = reg_df_multi
)

cat("\n--- Multivariate OLS: Secularization ~ All Predictors ---\n")
print(summary(model_multi))

# Multicollinearity diagnostics (condition number) referenced in report Section 6.2
X_design <- model.matrix(model_multi)
cat("\nCondition number of design matrix:", kappa(X_design), "\n")

# Coefficient plot (analogous to the Plotly scatter in the notebook)
coef_df <- as.data.frame(confint(model_multi))
names(coef_df) <- c("ci_low", "ci_high")
coef_df$variable <- rownames(coef_df)
```

```

coef_df$coef      <- coef(model_multi)[coef_df$variable]
coef_df <- coef_df %>% filter(variable != "(Intercept)")

ggplot(coef_df, aes(x = coef, y = variable)) +
  geom_point(size = 3, colour = "#2166ac") +
  geom_errorbarh(aes(xmin = ci_low, xmax = ci_high), height = 0.2, colour = "#2166ac") +
  geom_vline(xintercept = 0, linetype = "dashed", colour = "black") +
  labs(
    title = "Multivariate OLS - Coefficient Estimates (95% CI)",
    x      = "Estimated Effect on Secularization",
    y      = "Predictor"
  ) +
  theme_minimal(base_size = 12)

```

C. Ridge and Lasso Regression

```

# — 9d. RIDGE AND LASSO REGRESSION —————

X_mat <- model.matrix(secularization ~ fertility_rate + urbanization_rate +
                      life_expectancy + gdp_per_capita, data = reg_df_multi)[, -1]
y_vec <- reg_df_multi$secularization

set.seed(42)
cv_ridge <- cv.glmnet(X_mat, y_vec, alpha = 0) # alpha = 0 -> Ridge
set.seed(42)
cv_lasso <- cv.glmnet(X_mat, y_vec, alpha = 1) # alpha = 1 -> Lasso

cat("\nRidge: best lambda =", cv_ridge$lambda.min, "\n")
cat("\nLasso: best lambda =", cv_lasso$lambda.min, "\n")

ridge_coef <- as.matrix(coef(cv_ridge, s = "lambda.min"))
lasso_coef <- as.matrix(coef(cv_lasso, s = "lambda.min"))

reg_compare <- data.frame(
  variable = rownames(ridge_coef),
  OLS      = c(coef(model_multi)[ "(Intercept)" ], coef(model_multi)[ -1 ])[rownames(ridge_coef)],
  Ridge    = ridge_coef[, 1],
  Lasso    = lasso_coef[, 1]
)
cat("\nCoefficient comparison (OLS vs Ridge vs Lasso):\n")
print(reg_compare)

# Lasso path plot (shows coefficients shrinking to zero as lambda increases)
lasso_full <- glmnet(X_mat, y_vec, alpha = 1)
lasso_path_df <- as.data.frame(t(as.matrix(lasso_full$beta)))
lasso_path_df$log_lambda <- log(lasso_full$lambda)
lasso_path_long <- lasso_path_df %>%
  pivot_longer(-log_lambda, names_to = "variable", values_to = "coef")

ggplot(lasso_path_long, aes(x = log_lambda, y = coef, colour = variable)) +
  geom_line(linewidth = 1) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  labs(
    title = "Lasso Coefficient Paths",
    x      = "log(Lambda)",
    y      = "Coefficient",
    colour = "Predictor"
  )

```

```
) +  
theme_minimal(base_size = 12)  
  
# Compare predictive performance: Ridge/Lasso CV-MSE vs OLS  
ols_mse <- mean(resid(model_multi)^2)  
cat(sprintf("\nIn-sample MSE | OLS: %.3f | Ridge (CV-min): %.3f | Lasso (CV-min): %.3f\n",  
          ols_mse,  
          min(cv_ridge$cvm),  
          min(cv_lasso$cvm)))
```

D. Random Forest with Cross-Validation

```
# — 10. RANDOM FOREST REGRESSION —————  
  
set.seed(42)  
  
features <- c("fertility_rate", "urbanization_rate",  
             "life_expectancy", "gdp_per_capita")  
  
rf_df <- df %>%  
  select(secularization, all_of(features)) %>%  
  drop_na()  
  
# Train / test split (70 / 30)  
train_idx <- sample(seq_len(nrow(rf_df)), size = floor(0.7 * nrow(rf_df)))  
train_rf <- rf_df[ train_idx, ]  
test_rf <- rf_df[-train_idx, ]  
  
cat("\nTraining Random Forest (ntree = 300)... \n")  
rf_model <- randomForest(  
  secularization ~ .,  
  data = train_rf,  
  ntree = 300,  
  importance = TRUE  
)  
  
# — Evaluate —————  
y_pred_rf <- predict(rf_model, newdata = test_rf)  
r2_rf <- 1 - sum((test_rf$secularization - y_pred_rf)^2) /  
  sum((test_rf$secularization - mean(test_rf$secularization))^2)  
rmse_rf <- sqrt(mean((test_rf$secularization - y_pred_rf)^2))  
  
cat(sprintf("Random Forest | R² = %.4f | RMSE = %.4f\n", r2_rf, rmse_rf))
```

E. K-means Clustering

```
# — 11. K-MEANS CLUSTERING —————  
  
cluster_vars <- c("secularization", "fertility_rate",  
                 "urbanization_rate", "gdp_per_capita")
```

```

cluster_df_raw <- df %>%
  select(country_code, country_name, year, all_of(cluster_vars)) %>%
  drop_na()

# Standardise (z-scores) – essential for distance-based methods
X_scaled <- scale(cluster_df_raw %>% select(all_of(cluster_vars)))

## 11a. Elbow method to choose k
set.seed(42)
inertia <- map_dbl(2:7, function(k) {
  km <- kmeans(X_scaled, centers = k, nstart = 25, iter.max = 100)
  km$tot.withinss
})

elbow_df <- data.frame(k = 2:7, inertia = inertia)

ggplot(elbow_df, aes(x = k, y = inertia)) +
  geom_line(colour = "#2166ac", linewidth = 1) +
  geom_point(size = 3, colour = "#2166ac") +
  labs(
    title = "Elbow Method for Choosing Number of Clusters",
    x = "Number of Clusters (k)",
    y = "Total Within-Cluster SS (Inertia)"
  ) +
  theme_minimal(base_size = 12)

## 11b. Run k-means with k = 4
set.seed(42)
km_fit <- kmeans(X_scaled, centers = 4, nstart = 25, iter.max = 100)

cluster_df_raw$cluster <- km_fit$cluster

# IMPORTANT: kmeans() cluster IDs (1,2,3,4) are arbitrary and can vary between
# runs/seeds/package versions. Labels must be assigned FROM the centroid values,
# never hardcoded to a fixed ID, or labels can silently attach to the wrong cluster.
centroid_summary <- cluster_df_raw %>%
  group_by(cluster) %>%
  summarise(across(all_of(cluster_vars), mean), .groups = "drop")

label_lookup <- centroid_summary %>%
  mutate(cluster_label = case_when(
    gdp_per_capita == max(gdp_per_capita) ~ "Highly Developed / Secular",
    secularization == max(secularization) ~ "Secularizing Middle-Income",
    fertility_rate == max(fertility_rate) ~ "Developing Religious",
    TRUE ~ "Traditional / High Fertility"
  )) %>%
  select(cluster, cluster_label)

cluster_df_raw <- cluster_df_raw %>%
  left_join(label_lookup, by = "cluster")
cat("\nCluster summary (means):\n")
cluster_df_raw %>%
  group_by(cluster_label) %>%
  summarise(across(all_of(cluster_vars), ~ round(mean(.), 2)), .groups = "drop") %>%
  print()

cat("\nCluster sizes:\n")
print(table(cluster_df_raw$cluster_label))

```

```
## 11c. Visualise clusters

cluster_colours <- c(
  "Traditional / High Fertility" = "#6699CC",
  "Secularizing Middle-Income"   = "#E58A6E",
  "Developing Religious"         = "#5FAE8C",
  "Highly Developed / Secular"   = "#B79CD4"
)

# Centroids computed as ONE row per cluster (mirrors the Python version's
# single black X per cluster) – do NOT use stat_summary here, since combining
# it with point-level data can cause the centroid glyph to be drawn multiple
# times per cluster instead of once.
centroids_fert <- cluster_df_raw %>%
  group_by(cluster_label) %>%
  summarise(
    fertility_rate = mean(fertility_rate),
    secularization = mean(secularization),
    .groups = "drop"
  )

ggplot(cluster_df_raw,
  aes(x = fertility_rate, y = secularization, colour = cluster_label)) +
  geom_point(alpha = 0.55, size = 1.8) +
  geom_point(data = centroids_fert,
    aes(x = fertility_rate, y = secularization),
    inherit.aes = FALSE,
    shape = 4, size = 5, stroke = 1.5, colour = "black") +
  geom_text(data = centroids_fert,
    aes(x = fertility_rate, y = secularization, label = cluster_label),
    inherit.aes = FALSE,
    vjust = -1, size = 3.2, fontface = "bold", colour = "black") +
  scale_colour_manual(values = cluster_colours) +
  labs(
    title = "K-means Clusters: Fertility vs Secularization (k = 4)",
    x = "Fertility Rate",
    y = "% Unaffiliated (Secularization)",
    colour = "Cluster"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
    legend.text = element_text(size = 8))

# GDP (log) vs Secularization coloured by cluster
ggplot(cluster_df_raw,
  aes(x = gdp_per_capita, y = secularization, colour = cluster_label)) +
  geom_point(alpha = 0.55, size = 1.8) +
  scale_x_log10(labels = label_comma()) +
  scale_colour_manual(values = cluster_colours) +
  labs(
    title = "K-means Clusters: GDP per Capita vs Secularization (k = 4)",
    x = "GDP per Capita (log scale, USD)",
    y = "% Unaffiliated (Secularization)",
    colour = "Cluster"
  ) +
  theme_minimal(base_size = 12) +
  theme(legend.position = "bottom",
    legend.text = element_text(size = 8))
```

F. Hierarchical Clustering

```
# — 11d. HIERARCHICAL CLUSTERING (VALIDATES K-MEANS SOLUTION) —————
# If the same 4-group structure emerges from a totally different clustering
# algorithm, that is strong evidence the groups reflect real data structure
# rather than an artifact of k-means' specific assumptions.

dist_mat <- dist(X_scaled, method = "euclidean")
hc_fit   <- hclust(dist_mat, method = "ward.D2")

# Dendrogram (subsample for readability since n is in the hundreds)
set.seed(42)
sample_idx_hc <- sample(seq_len(nrow(X_scaled)), min(80, nrow(X_scaled)))
hc_sub <- hclust(dist(X_scaled[sample_idx_hc, ]), method = "ward.D2")

plot(hc_sub,
     labels = FALSE,
     main   = "Hierarchical Clustering Dendrogram (Ward's method, n = 80 sample)",
     xlab   = "Countries (sample)",
     ylab   = "Height")
rect.hclust(hc_sub, k = 4, border = "#d6604d")

# Cut full tree at k = 4 and compare against k-means labels
hc_clusters_4 <- cutree(hc_fit, k = 4)
agreement_tab <- table(KMeans = cluster_df_raw$cluster_label, Hierarchical = hc_clusters_4)
cat("\nAgreement between K-means and Hierarchical clustering (k = 4):\n")
print(agreement_tab)
```